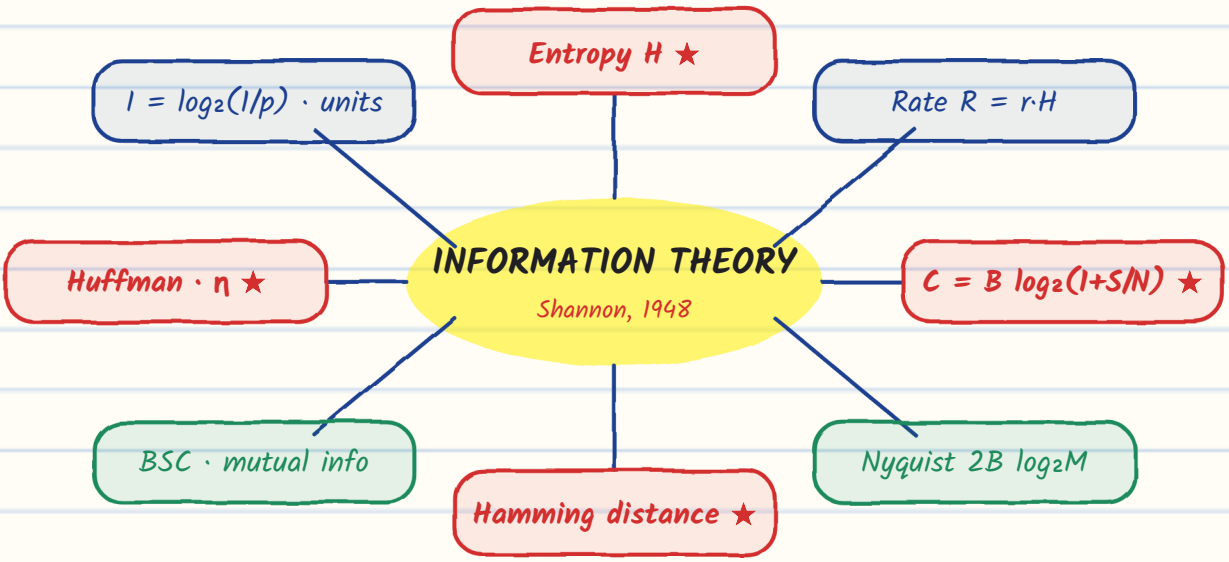
Information Theory

Entropy · Rate · Huffman · Shannon · Capacity: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = method / applied

➜ Syllabus coverage

Communication Theory: "Information theory: entropy, information rate, Shannon's theorem, channel capacity". Also: source coding (Shannon–Fano, Huffman), the BSC, the Nyquist limit and error-control basics. Sampling is in the Spectral Analysis & Sampling notes; PCM, modulation and multiplexing in the Modulation, Pulse Signals & Multiple Access notes.

High-yield list

① $I = \log_2(1/p)$
② $H_{max} = \log_2 M$
③ $R = r \cdot H$
④ Huffman, $\eta = H/L$
⑤ $C = B \log_2(1 + S/N)$
⑥ limit −1.6 dB
⑦ d_{min} rules

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

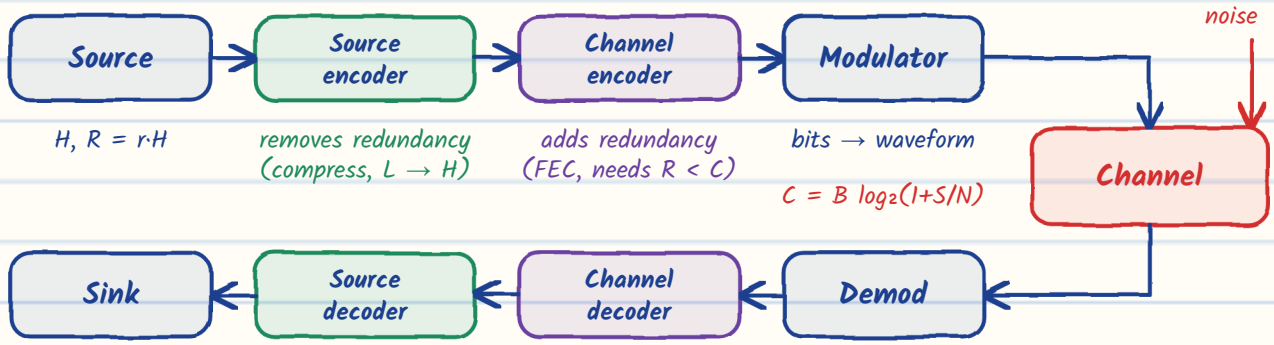
➜ Contents

✓ System model, information ... p. 2
✓ Entropy & information rate ... p. 3
✓ Source coding, Shannon–Fano ... p. 4
✓ Huffman coding ..... p. 5
✓ BSC & coding theorem ..... p. 6
✓ Shannon–Hartley & dB ..... p. 7
✓ Shannon limit, trade-off, Nyquist ... p. 8
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1 The Digital Communication System

MCQ + Written

Definition Information theory is the mathematics of measuring, compressing and reliably transmitting information. Founded by Claude E. Shannon in "A Mathematical Theory of Communication" (Bell System Technical Journal, 1948): the "father of information theory".



Shannon's model: source coding squeezes the redundancy out, channel coding puts controlled redundancy back in

Memory trick "**S**ource coding **S**hrinks, **C**hannel coding **C**overs." Source coding = compression (Huffman, ZIP: L, η); channel coding = protection (parity, Hamming, CRC: code rate k/n, d_{min}).

2 Measure of Information

high-yield

Self-information A message of probability p carries $I = \log_2(1/p) = -\log_2 p$ bits. The less likely the message, the more information it carries ("sun rose today" ≈ 0 bits; "earthquake in Dhaka" = many bits).

- $p = 1$ (certain event) → $I = 0$; always $I \geq 0$
- $p_1 < p_2 \rightarrow I_1 > I_2$
- Independent messages: $I(x_1x_2) = I(x_1) + I(x_2)$ (log turns × into +)
- Binary digit with $p = ½$ → exactly **1 bit**

Worked example ✍️ $p = 1/8 \rightarrow I = \log_2 8 =$ **3 bits**. A card from a 52-card deck: $\log_2 52 =$ **5.70 bits**; "it is an ace" ($p = 4/52 = 1/13$): $\log_2 13 =$ **3.70 bits**.

Log base	Unit	= bits
2	**bit** (shannon)	1
e	**nat**	1.443
10	**decit / Hartley**	3.322

Exam trap **Baud** is a unit of **symbol rate**, not of information. And $\log_2 x = 3.322 \log_{10} x$: keep that conversion for calculator-free MCQs.

3 Entropy H: Average Information per Symbol high-yield

Definition For a discrete memoryless source (DMS) with M symbols of probabilities $p_1 \dots p_M$: $H(X) = \Sigma\, p_i \log_2(1/p_i)$ bits/symbol. Entropy = average self-information = average uncertainty removed per symbol = the limit of lossless compression.

Bounds

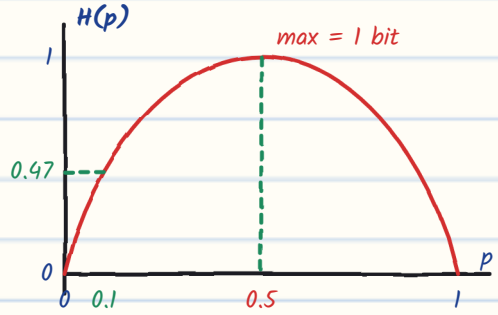
$0 \le H \le \log_2 M$

$H = 0 \rightarrow$ one symbol certain ($p = 1$)

$H = \log_2 M \rightarrow$ all **equiprobable** ($p = 1/M$)

n-th extension: $H(X^n) = n \cdot H(X)$

Binary source: $H(p)$ max **1 bit at $p = 0.5$**



binary entropy function

Source	Probabilities	H (bits/symbol)
Fair coin	½, ½	**1**
Biased coin	0.9, 0.1	0.469
4 equal symbols	¼ each	**2** = $\log_2 4$
Dyadic source	½, ¼, ⅛, ⅛	**1.75**
Fair die	1/6 each	2.585 = $\log_2 6$
English letters	letter frequencies	≈ 4.1 (max $\log_2 26$ = 4.70)

Worked example p = ½, ¼, ⅛, ⅛: H = ½(1) + ¼(2) + ⅛(3) + ⅛(3) = 0.5 + 0.5 + 0.375 + 0.375 = **1.75 bits/symbol** (< $\log_2 4 = 2$, since not equiprobable). Source redundancy = $1 - H/H_{max} = 1 - 1.75/2 = 12.5\%$.

4 Information Rate R

Definition Source emitting r symbols/s: $R = r \cdot H$ bits/s.

Worked example A 4 kHz signal sampled at the Nyquist rate (r = 8000 samples/s), quantized to 4 levels with p = ⅛, ⅜, ⅜, ⅛: $H = 2 \times \frac{1}{8} \times 3 + 2 \times \frac{3}{8} \times \log_2(8/3) = 0.75 + 1.061 = 1.811$ **bits** $\rightarrow R = 8000 \times 1.811 \approx$ **14.5 kbit/s** (equiprobable levels: 16 kbit/s).

Memory trick "**Equal → log, else less**": any imbalance lowers H.

Exam trap H = 2 bits/symbol, r = 2000/s → R = **4000 bit/s**, not 1000.

5 Source Coding: Length, Efficiency, Kraft

high-yield

Goal Make the average codeword length L as small as possible without losing information: **short codes for frequent symbols** (the Morse idea).

Formulas

Average length: $L = \sum p_i l_i$ bits/symbol

Efficiency: $\eta = H / L$ (binary code)

Redundancy: $\gamma = 1 - \eta$

D-ary code: $\eta = H / (L \log_2 D)$

Kraft (prefix): $\sum 2^{-l_i} \leq 1$

Fixed-length code: $l = \lceil \log_2 M \rceil$ bits

Shannon's 1st theorem (source coding) For any uniquely decodable code $L \geq H$; a prefix code with $H \leq L < H + 1$ always exists, and coding long blocks of symbols brings L/symbol as close to H as we like. $\eta = 100\%$ only when every p_i is a power of ½ (dyadic).

Code	A	B	C	D	Verdict
I	0	10	110	111	**prefix (instantaneous)**: no codeword starts another
II	0	01	011	111	not prefix (0 starts 01), still uniquely decodable
III	0	1	01	10	**ambiguous**: "01" = AB or C; Kraft sum = 1.5 > 1

6 Shannon–Fano Coding (top-down)

1. List the symbols in **decreasing** probability
2. **Split** the list into two parts with totals as **nearly equal** as possible; give 0 to the upper part, 1 to the lower
3. Repeat inside each part until every part holds one symbol

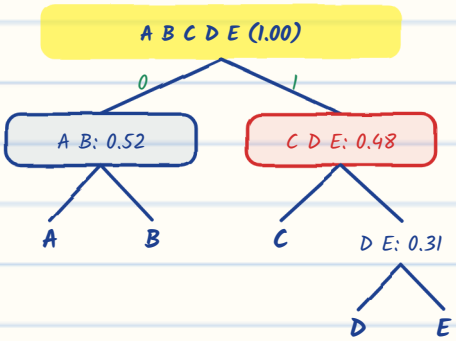
Worked example ✍️ A 0.35, B 0.17, C 0.17, D 0.16, E 0.15

Symbol	A	B	C	D	E
Code	00	01	10	110	111
l	2	2	2	3	3

Splits: 0.52 | 0.48, then 0.17 | 0.31

$L = 2(0.35 + 0.17 + 0.17) + 3(0.16 + 0.15) =$ **2.31 bits**

$H = 2.233$ bits → $\eta = 2.233/2.31 =$ **96.7%**, $\gamma =$ 3.3%



split as evenly as possible, top-down

Exam trap Shannon–Fano is **not always optimal**: on this same source Huffman gives 2.30 bits (next sheet).

7 Huffman Coding (bottom-up, optimal)

written favourite

Definition D. A. Huffman (MIT, 1952). Builds the code tree from the leaves up by repeatedly merging the two least probable entries. It gives the minimum average length of any symbol-by-symbol prefix code.

1. Arrange the probabilities in decreasing order
2. **Merge the two smallest** into one node whose probability is their sum; label the branches 0 and 1
3. Re-sort and repeat until one node of probability 1 remains (M − 1 merges)
4. Read each codeword from the root down to the leaf

Worked example ✍️ Same source: A 0.35, B 0.17, C 0.17, D 0.16, E 0.15

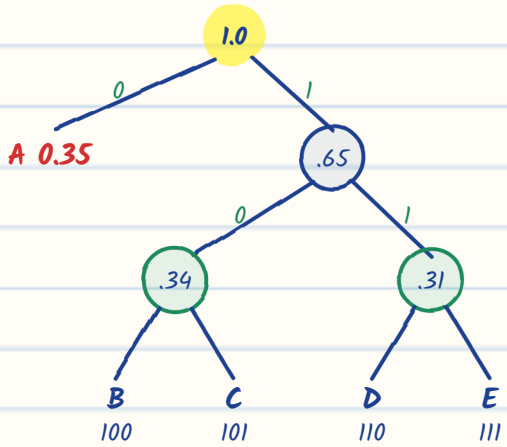
1. D + E = 0.15 + 0.16 = **0.31**
2. B + C = 0.17 + 0.17 = **0.34**
3. 0.31 + 0.34 = **0.65**
4. 0.65 + A 0.35 = **1.00**

Codes: A = **0**, B = 100, C = 101, D = 110, E = 111

$L = 0.35 \times 1 + 0.65 \times 3 =$ **2.30 bits**

$\eta = 2.233/2.30 =$ **97.1%**, $\gamma = 2.9\%$.

Kraft: $\frac{1}{2} + 4 \times \frac{1}{8} = 1$ ✓



left branch = 0, right branch = 1

Code for A–E	L (bits)	η	Remark
Fixed length	3	74.4%	$\lceil \log_2 5 \rceil = 3$
Shannon–Fano	2.31	96.7%	top-down split
Huffman	**2.30**	**97.1%**	optimal; H = 2.233 is the floor

Memory trick **Huffman = Hug the two smallest** (bottom-up merge). **Fano = Fold in half** (top-down split).

Exam trap Ties can give **different codewords** (and code-length variance), but the average length L is always the same. Dyadic source ½, ¼, ⅛, ⅛ → codes 0, 10, 110, 111, L = H = 1.75, η = **100%**.

IT link Huffman (lossless) is the last stage of DEFLATE (ZIP, gzip, PNG = LZ77 + Huffman), of JPEG and of MP3 (their loss comes from earlier stages). Entropy explains why random or already-compressed data (H ≈ 8 bits per byte) cannot be zipped further.

8 Joint & Conditional Entropy, Mutual Information

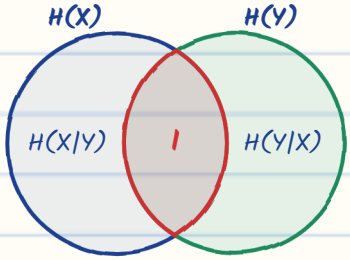
Joint: $H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$

Equivocation $H(X|Y)$: input uncertainty left after seeing the output (lost to noise)

Mutual information (what gets through):

$I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$

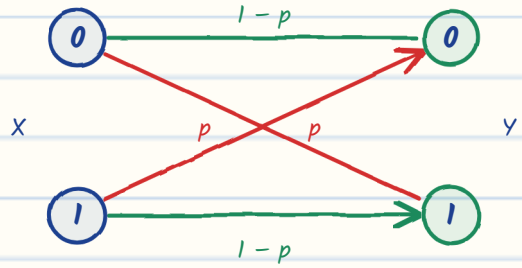
- $I \geq 0$, symmetric · independent X, Y → $I = 0$
- Noiseless channel → $H(X|Y) = 0$, $I = H(X)$



the information diagram

9 Channel Models & Channel Capacity high-yield

Definition A **discrete memoryless channel** (DMC) is described by its transition matrix $P(y_j|x_i)$. Its capacity is the largest mutual information over all input distributions: $C = \max I(X;Y)$ bits per channel use.



binary symmetric channel (BSC), crossover p

BSC capacity (equiprobable input is optimal):

$C = 1 - H(p)$ bits/use

p	0	0.01	0.1	0.2	**0.5**	1
C	1	0.919	0.531	0.278	**0**	1

Noiseless, M symbols: $C = \log_2 M$ · erasure channel: $C = 1 - \alpha$

Exam trap BSC with $p = 1$ is **perfect** ($C = 1$: just invert every bit). The useless one is **$p = 0.5$**.

10 Shannon's Channel Coding Theorem high-yield

Shannon's 2nd theorem If the information rate $R < C$, there exists a channel code that makes the probability of error arbitrarily small. ($R = C$ is the boundary.) If $R > C$, no code can do so: the error probability stays bounded away from zero.

Memory trick "**Below C, code and be free; above C, errors you'll see.**" Good codes **exist** (turbo, LDPC, polar come close).

Exam trap "Error-free" means **arbitrarily low error probability**, not zero errors; the price is **long codes / delay**.

11 Shannon–Hartley Capacity Theorem

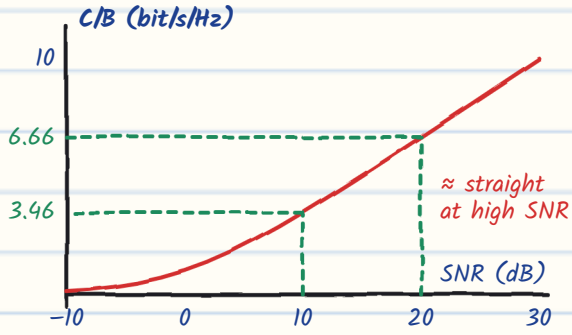
high-yield

Statement For a band-limited channel of bandwidth B (Hz) with additive white Gaussian noise (**AWGN**) and average signal-to-noise power ratio S/N:

$$C = B \log_2(1 + S/N) \text{ bits/s}$$

where $N = N_0 B$ (N_0 = noise power spectral density).

- S/N is a linear power ratio, never dB
- C grows **linearly** with B, only **logarithmically** with S/N
- High S/N: +3 dB (×2) adds ≈ 1 bit/s/Hz
- Quick estimate: $C \approx B \times SNR_{dB}/3$
- S/N = 0 → C = 0; S/N → ∞ (fixed B) → C → ∞
- An upper limit: real modems stay below it



$C/B = \log_2(1 + S/N)$ against SNR in dB

dB	S/N	$\log_2(1+S/N)$
0	1	1
3	≈ 2	1.58
4.77	3	2
10	10	3.46
11.76	15	4
17.99	63	6
20	100	6.66
30	**1000**	**9.97**

dB conversions

Power: $SNR_{dB} = 10 \log_{10}(S/N)$

Voltage: $20 \log_{10}(V_s/V_n)$

Back to linear: $S/N = 10^{SNR_{dB}/10}$

Every +10 dB = ×10 power; +3 dB ≈ ×2

Memory trick Pick $S/N = 2^k - 1$ (1, 3, 7, 15, 31, 63, 255) and $\log_2(1 + S/N) = k$ exactly: that is how MCQ numbers are built.

Worked examples ✍

1. B = 4 kHz, S/N = 15: $C = 4000 \times \log_2 16 = 4000 \times 4 =$ **16 kbit/s**
2. Telephone line B = 3 kHz, SNR = 30 dB → S/N = 1000: $C = 3000 \times \log_2 1001 = 3000 \times 9.97 \approx$ **29.9 kbit/s** (estimate: 3000 × 30/3 = 30 k ✓)
3. B = 1 MHz: 10 dB → 3.46 Mbit/s; 20 dB → 6.66 Mbit/s. B = 10 MHz at 10 dB → **34.6 Mbit/s** (10× B → 10× C)
4. Minimum S/N to carry 20 kbit/s in 5 kHz: $\log_2(1 + S/N) = 4$ → S/N = 15 = **11.76 dB**

Exam trap Plugging "20 dB" as 20 gives $\log_2 21 = 4.39$ instead of $\log_2 101 = 6.66$. **Convert dB to linear first.**

12 Infinite Bandwidth & the Shannon Limit high-yield

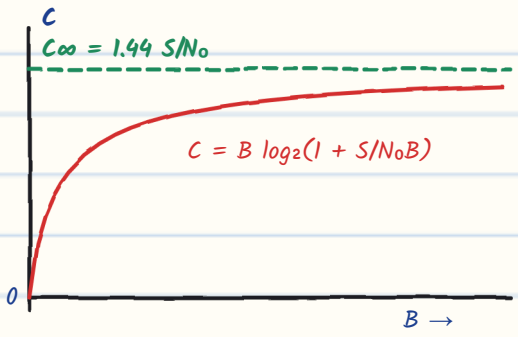
Fixed S and N_0: wider B lets in more noise ($N = N_0B$). As $B \to \infty$, with $\ln(1 + x) \approx x$:

$C_\infty = (S/N_0) \log_2 e \approx 1.44\ S/N_0$

Capacity **saturates**: infinite B ≠ infinite C.

Put $S = E_b \cdot C$: $E_b/N_0 \geq \ln 2 = 0.693 =$

$-1.59 \approx -1.6$ dB



capacity flattens out as bandwidth grows

Minimum $E_b/N_0 = (2^{C/B} - 1)/(C/B)$: **0 dB** at 1 bit/s/Hz, 1.76 dB at 2, 5.74 dB at 4.

Exam trap −1.6 dB limits E_b/N_0 (energy per bit), not SNR, and needs infinite bandwidth. Below it, reliable communication is impossible at **any** rate.

13 Bandwidth–SNR Trade-off

Same C = 12 kbit/s			
B	S/N	dB	S (× N_0·1 kHz)
3 kHz	15	11.76	45
4 kHz	7	8.45	28
6 kHz	3	4.77	18
12 kHz	1	0	12
∞	→ 0	−∞	**8.32 floor**

For a fixed C, bandwidth can be traded for signal power: more B → less S/N (and less S) needed. Basis of **spread spectrum / CDMA** and wideband FM.

Wide B, low S/N: deep space, CDMA. Narrow B, high S/N: phone modems, high-order QAM.

14 Nyquist Capacity (noiseless channel)

Nyquist bit rate A noiseless channel of bandwidth B carries at most 2B symbols/s (baud); with M levels: $C = 2B \log_2 M$ bits/s. Shannon sets the noise ceiling; Nyquist says how many levels reach it.

Worked examples ✍ ① B = 3 kHz: M = 2 → 6 kbit/s; M = 4 → **12 kbit/s**. ② B = 1 MHz, S/N = 63: Shannon $C = 10^6 \times \log_2 64 =$ **6 Mbit/s**; use 4 Mbit/s for margin → $2 \times 10^6 \times \log_2 M = 4 \times 10^6$ → **M = 4**. ③ 265 kbit/s on a noiseless 20 kHz line: $\log_2 M = 6.625$ → M = 98.7 → use **128** levels.

Memory trick **Nyquist = No noise, Number of levels** ($2B \log_2 M$). **Shannon = S/N** ($B \log_2(1 + S/N)$). Noisy channel: the answer is the **smaller** of the two.

15 Error-Control (Channel) Coding Basics

MCQ + Written

Idea The channel coding theorem says reliable transmission needs **controlled redundancy**. An **(n, k) block code** maps k data bits to n-bit codewords: $n - k$ check bits, **code rate $r = k/n$** (< 1).

	ARQ (Automatic Repeat reQuest)	FEC (Forward Error Correction)
Receiver	detects, asks for retransmission	detects **and corrects** itself
Used in	TCP, Wi-Fi MAC, CRC + retransmit	satellite, deep space, CD/DVD, 4G/5G

➜ Parity check (single-bit)

Even parity: total 1s (data + parity) even. 1011001 has four 1s → even parity bit **0**, odd parity bit **1**. Detects any **odd** number of errors, misses even numbers, **cannot correct** ($d_{min} = 2$).

➜ Hamming distance

$d(a, b)$ = positions that differ = 1s in $a \oplus b$. $10101 \oplus 11110 = 01011 \rightarrow$ **$d = 3$** d_{min} = smallest distance between any two codewords.

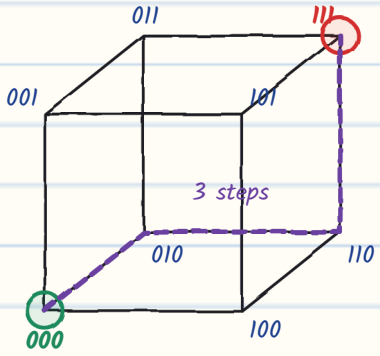
d_{min} rules

Detect up to **$s = d_{min} - 1$** errors

Correct up to **$t = \lfloor (d_{min} - 1)/2 \rfloor$** errors

To correct t: $d_{min} \geq 2t + 1$ · detect s: $d_{min} \geq s + 1$

d_{min}	2	3	4	5	7
detect	1	2	3	4	6
correct	0	1	1	2	3



repetition code {000, 111}: $d_{min} = 3$ corrects 1 error (majority vote)

➜ Hamming code (single-error correcting, $d_{min} = 3$)

Check bits r: **$2^r \geq m + r + 1$** (m data bits): $m = 4 \rightarrow r = 3$, **(7, 4)**; $m = 7$ (ASCII) $\rightarrow r = 4$, (11, 7). Parity bits at positions **1, 2, 4, 8 …**; P1 checks 1, 3, 5, 7 · P2: 2, 3, 6, 7 · P4: 4, 5, 6, 7.

Worked example ✍ Data 1011 (even parity) → positions 3, 5, 6, 7 = 1, 0, 1, 1. $P1 = 1 \oplus 0 \oplus 1 = 0$ · $P2 = 1 \oplus 1 \oplus 1 = 1$ · $P4 = 0 \oplus 1 \oplus 1 = 0$ → codeword **0110011**

Received 0110001: checks C4 C2 C1 = 1 1 0 → syndrome **110 = 6** → flip bit 6 → 0110011 ✓

Memory trick "**Detect d − 1, correct half of that**" (round down).

Exam trap Parity **cannot correct** even one error; CRC only **detects**.

Night-Before-the-Exam Sheet

Formula sheet

$I = \log_2(1/p)$ bits · $p = 1 \to I = 0$

$H = \sum p \log_2(1/p)$ · $0 \le H \le \log_2 M$

$R = r \cdot H$ bits/s · $H(X^n) = nH$

Kraft $\sum 2^{-l} \le 1$ · $H \le L < H + 1$

BSC: $C = 1 - H(p)$ · $p = 0.5 \to C = 0$

$SNR_{dB} = 10 \log_{10}(S/N)$ · 30 dB = 1000

Nyquist (noiseless): $C = 2B \log_2 M$

Detect $d_{min} - 1$ · correct $\lfloor (d_{min} - 1)/2 \rfloor$

1 nat = 1.443 bit · 1 Hartley = 3.322 bit

Binary: $H(0.5) = 1$ bit · $H(0.1) = 0.469$

$L = \sum p l$ · $\eta = H/L$ · $\gamma = 1 - \eta$

$I(X;Y) = H(X) - H(X|Y)$ · $C = \max I$

$C = B \log_2(1 + S/N)$, S/N linear

$C_\infty = 1.44\ S/N_0$ · $E_b/N_0 \ge -1.6$ dB

Hamming: $2^r \ge m + r + 1$ · rate k/n

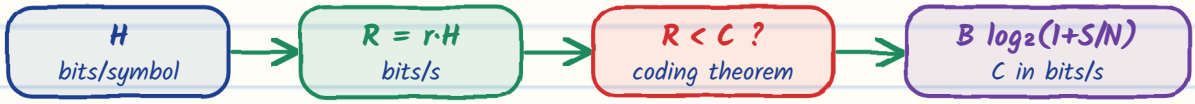
Noisy channel: min(Shannon, Nyquist)

→ Top 12 one-liners for MCQs

1. Father of information theory: Shannon, 1948
2. Base-2 unit = bit; base e = nat; base 10 = decit
3. $p = 1/8 \to 3$ bits; $p = 1 \to 0$ bits
4. 4 equiprobable symbols → H = 2 bits
5. ½, ¼, ⅛, ⅛ → H = 1.75 bits
6. Huffman (1952): optimal prefix code, merge 2 smallest
7. Shannon–Fano: top-down, not always optimal
8. B = 4 kHz, S/N = 15 → C = 16 kbit/s
9. 30 dB = 1000; 20 dB = 100; 3 dB ≈ 2
10. $B \to \infty$: $C \to 1.44\ S/N_0$ (finite)
11. Parity detects odd errors only, corrects none
12. Hamming (7, 4): 3 parity bits, corrects 1

Top traps dB into the formula without converting · H in bits/symbol vs R in bits/s · baud ≠ bit/s · BSC p = 1 is perfect, p = 0.5 is useless · "error-free" = arbitrarily small error, only for R < C · infinite B ≠ infinite C · Huffman codewords not unique, L is · noisy channel: min(Shannon, Nyquist)

All tricks together Source coding Shrinks, Channel coding Covers · equal → log, else less · Huffman hugs the two smallest, Fano folds in half · $S/N = 2^k - 1$ gives k · Nyquist = No noise, Number of levels · below C, code and be free · detect d − 1, correct half of that



the four syllabus words in one chain

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. Unit of information for log base e:
(a) bit (b) decit
(c) nat (d) baud

2. Entropy of 8 equiprobable symbols:
(a) 3 bits (b) 8 bits
(c) 1 bit (d) 2.5 bits

3. Information in an event of $p = 1/16$:
(a) 16 bits (b) 0.0625 bit
(c) 2 bits (d) 4 bits

4. Capacity of a BSC with $p = 0.5$:
(a) 1 bit/use (b) 0
(c) 0.5 bit/use (d) infinite

5. $B = 3$ kHz, $S/N = 7$ (linear). C = ?
(a) 21 kbit/s (b) 24 kbit/s
(c) 9 kbit/s (d) 3 kbit/s

6. 20 dB SNR is a power ratio of:
(a) 20 (b) 200
(c) 10 (d) 100

7. Shannon limit on E_b/N_0:
(a) 0 dB (b) −1.6 dB
(c) 1.6 dB (d) 3 dB

8. $d_{min} = 5$ corrects up to:
(a) 2 errors (b) 4 errors
(c) 5 errors (d) 1 error

9. Huffman coding gives a … code:
(a) lossy (b) fixed-length
(c) lossless prefix (d) error-correcting

10. Noiseless 4 kHz, 8 levels: max rate?
(a) 24 kbit/s (b) 12 kbit/s
(c) 32 kbit/s (d) 8 kbit/s

Answer key → 1-c · 2-a · 3-d · 4-b · 5-c · 6-d · 7-b · 8-a · 9-c · 10-a

Written question A telephone channel has a bandwidth of 3.4 kHz and an SNR of 30 dB. (a) Find its Shannon capacity. (b) What SNR would be needed to carry 40 kbit/s? (c) How many signal levels should a Nyquist-type modem use to stay within capacity? (d) State the spectral efficiency. 8 marks

1. $S/N = 10^{30/10} = 1000$; $C = 3400 \times \log_2 1001 = 3400 \times 9.967 \approx$ **33.9 kbit/s** — 3
2. $\log_2(1 + S/N) = 40\,000/3400 = 11.76 \rightarrow S/N = 2^{11.76} - 1 \approx 3479 \rightarrow$ **≈ 35.4 dB** — 2
3. $2B \log_2 M \le C \rightarrow \log_2 M \le 33\,900/6800 = 4.98$. $M = 32$ needs 34.0 kbit/s > C, so use **$M = 16$** (27.2 kbit/s) — 2
4. $C/B \approx 9.97$ bit/s/Hz ($\approx SNR_{dB}/3 = 10$ ✓) — 1

→ Before the exam ✓

- ✓ Can compute I, H and R with units
- ✓ Can build Shannon–Fano and Huffman codes, find L, η, γ
- ✓ Can convert dB ↔ linear and apply Shannon–Hartley
- ✓ Can use d_{min} rules and encode Hamming (7, 4)

Also practise:
① Huffman: 0.4, 0.2, 0.2, 0.1, 0.1
② Derive $C_\infty = 1.44\ S/N_0$
③ BSC capacity, $p = 0.1$
④ Hamming (7, 4) of 1101